

Ψ -Risk-DT: A Neurosymbolic Framework for Adaptive Zero-Day Threat Detection in Digital Twin Ecosystems — Additions

TABLE I
STRUCTURAL ENTROPIC POINTS AND THEIR ROLE IN THE DT OPERATIONAL CYCLE.

Point	Phase	Meaning
ΔH_{acq}	Acquisition	Informational loss due to sensor noise, packet loss, and measurement distortion
ΔH_{sync}	Alignment	Temporal staleness between the physical system and its virtual replica
ΔH_{sim}	Mismatch	Simulation–reality divergence; formally linked to $\Delta_t = O_t - E_t$
ΔH_{risk}	Neural scoring	Classification uncertainty (epistemic uncertainty of the ARNN output)
ΔH_{ctrl}	Control	Decision–actuation delay; degradation of response effectiveness due to latency

I. DUMMY SECTION

Text referencing MSU and conclusion for label resolution.

A. Distributed Entropy as a Structural Signal

In standard IDS pipelines, entropy is computed as a single aggregate metric over a traffic window. However, in Digital Twin ecosystems uncertainty does not emerge at a single point: it is *distributed* across the entire observation–inference–actuation cycle. Sensor noise degrades input quality before analysis begins; stale synchronization inflates mismatch even without an attack; classifier confidence varies independently of traffic entropy; and actuation delays erode the effectiveness of correct decisions.

We therefore decompose the entropy deviation into five *structural entropic points*, each capturing a distinct source of epistemic uncertainty within the DT lifecycle:

$$\Delta \mathbf{H}_{\text{struct}} = (\Delta H_{\text{acq}}, \Delta H_{\text{sync}}, \Delta H_{\text{sim}}, \Delta H_{\text{risk}}, \Delta H_{\text{ctrl}}). \quad (1)$$

Table I defines each component and maps it to the corresponding phase of the Ψ -Risk-DT operational cycle (visualized in Fig. 1).

This decomposition carries two implications for the Ψ -Risk-DT architecture:

- 1) **Multi-entropic gating.** The operator Ψ can be generalized to accept the vectorial input $\Delta \mathbf{H}_{\text{struct}}$ rather than a scalar ΔH_t . Each entropic point could trigger independent MSU updates on the corresponding DT component (e.g., ΔH_{acq} on sensor subgraphs, ΔH_{sim} on simulation subgraphs), aligning naturally with MSU’s modular design (Section I).
- 2) **Diagnostic observability.** The vector $\Delta \mathbf{H}_{\text{struct}}$ provides an *entropy map* of the DT lifecycle: not only *how much* uncertainty is present, but *where* it originates. This enables targeted hardening (e.g., improving sensor calibration when ΔH_{acq} dominates) and more informative explanations in the Ψ -UI.

In the current implementation, ΔH_t aggregates these components implicitly through Shannon entropy over mixed feature sets. The explicit vectorial formulation in Eq. (1) opens a direct path toward finer-grained governance and constitutes a primary direction for future refinement, as discussed in Section I.

More broadly, the trajectory from H-DIR (semantic interoperability) through (H-DIR)² (entropic risk detection) to Ψ -Risk-DT (neurosymbolic governance) suggests a convergent path toward a *Unified Entropy-Governed Digital Twin (UEG-DT) Theory*, in which entropy serves as the overarching regulatory principle for state evolution, knowledge certification, and adaptive control. The structural entropic decomposition introduced in Section I-A provides a concrete formal basis for this unification: rather than a single detection trigger, entropy becomes a *distributed governance signal* across the entire DT lifecycle. Formalizing UEG-DT as a compositional framework—where serialization, entropic selection, neural scoring, cyber-physical mismatch detection, and neurosymbolic governance are composed as

$$\Psi \circ f_{\text{ARNN}} \circ g(\Delta_t) \circ \Delta H \circ f_{\text{RDF}} : \Omega \longrightarrow (S_{t+1}, KB_{t+1}, u_t), \quad (2)$$

where Ω denotes the raw telemetry space, $g(\Delta_t)$ the cyber-physical mismatch function, and ΔH the distributed entropic operator from Eq. (1)—constitutes the primary theoretical objective of forthcoming work. This compositional view positions the three existing contributions as stages of a single, entropy-governed cognitive architecture for Digital Twins.

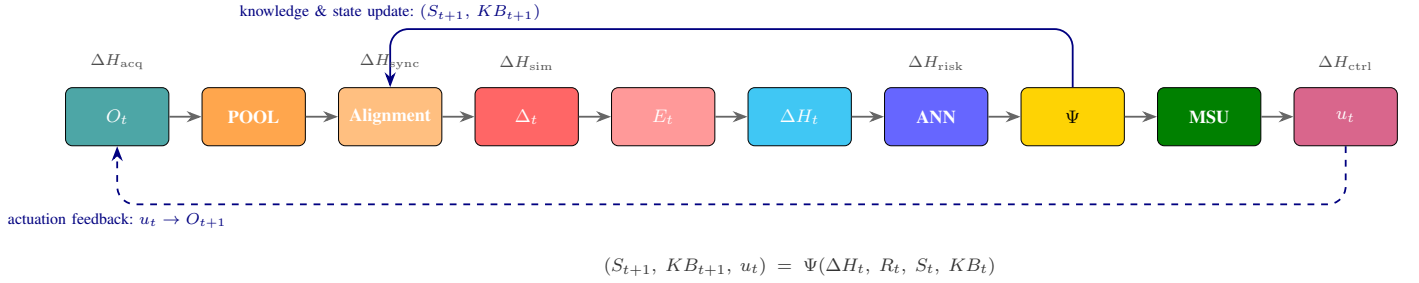


Fig. 1. Closed-loop operational flow of Ψ -Risk-DT. The main chain (left to right) processes observations O_t through pooling, alignment, mismatch detection, entropic filtering, neural scoring, neurosymbolic governance (Ψ), and modular semantic update (MSU), producing a control action u_t . Two feedback paths close the loop: an upper path updates the knowledge base and system state; a lower path propagates actuation effects to the next observation cycle. The five structural entropic points (ΔH_{acq} through ΔH_{ctrl} , cf. Table I) are annotated above the corresponding stages.